Limited-Resource Adapters Are Regularizers, Not Linguists

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Abstract

Cross-lingual transfer from related high-resource languages is a well-established strategy to enhance low-resource language technologies. Prior work has shown that adapters show promise for, e.g., improving low-resource machine translation (MT). In this work, we investigate an adapter souping method combined with cross-attention fine-tuning of a pre-trained MT model to leverage language transfer for three low-resource Creole languages, which exhibit relatedness to different language groups across distinct linguistic dimensions. Our approach improves performance substantially over baselines. However, we find that linguistic relatedness—or even a lack thereof—does not covary meaningfully with adapter performance. Surprisingly, our cross-attention fine-tuning approach appears equally effective with randomly initialized adapters, implying that the benefit of adapters in this setting lies in parameter regularization, and not in meaningful information transfer. We provide analysis supporting this regularization hypothesis. Our findings underscore the reality that neural language processing involves many success factors, and that not all neural methods leverage linguistic knowledge in intuitive ways.

1 Introduction and Background

For a majority of the world's languages, data scarcity remains a barrier for reliable machine translation (MT). Cross-lingual transfer from high-resource to low-resource languages has been shown to help mitigate this (Zoph et al., 2016; Tars et al., 2021), but has still not closed the gap between high- and low-resource scenarios. In this work, we seek to exploit methods from parameter-efficient fine-tuning (PEFT) to further improve cross-lingual transfer for a Creoles, an under-served group of low-resource languages.

PEFT methods such as bottleneck adapters (Houlsby et al., 2019; Pfeiffer et al., 2020), LoRA (Hu et al., 2021), and $(IA)^3$ (Liu et al., 2022a) are widely used to adapt language models to low-resource languages (Parović et al., 2022; Parovic et al., 2023; Rathore et al., 2023; Vázquez et al., 2024; Dementieva et al., 2025). This is in part because they are robust to hyperparameter choices and training data amounts, in contrast to full model fine-tuning (Pfeiffer et al., 2024). Some PEFT methods can successfully use monolingual data to improve MT performance; for example, Üstün et al. (2021a) train denoising adapters for both source and target languages, inserting the adapters between the encoder and decoder layers of an MT model, respectively. By freezing all model weights except for the cross-attention of the decoder, they then carry out cross-attention fine-tuning (CA-FT) with only a limited amount of parallel data. Chronopoulou et al. (2023a) also address data scarcity in adaptation to new domains following

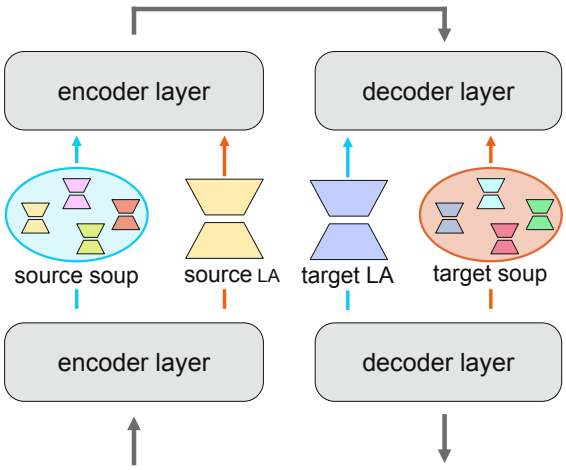


Figure 1: Overview of the MT transfer experiments between English and Creole languages. ↑ path shows the use of source language adapter in the encoder and the 'souped' (weight-averaged) target and transfer language adapters in the decoder layers. ↑ path shows the opposite direction.

Wortsman et al. (2022) by "souping" (*i.e.,* weight-space averaging) pretrained domain adapters, as in Equation 1 (where θ_i represent the weights of each domain adapter). We combine the principles of these prior works in our paper.

$$\theta_{soup} = \frac{1}{l}\sum_{i=1}^{l}\theta_i \quad (1)$$

Prior work has also catalyzed transfer by training language family adapters to leverage phylogenetic relations between languages (Faisal and Anastasopoulos, 2022; Chronopoulou et al., 2023b). Other factors of selecting transfer languages have been investigated in the past besides phylogeny. These include similarity in terms of explicit grammatical features like word order (Pires et al., 2019), similarity in terms of language representation space (Üstün et al., 2022), as well as lexical overlap (Pfeiffer et al., 2021; de Vries et al., 2022). Other metrics have been proposed to select candidate transfer languages, including Subword Evenness (SuE; Pelloni et al., 2022) which quantifies how uneven subword lengths are in a multilingual tokenizer: uneven implies successful transfer.

In this work we focus on MT for Creole languages, a set of low-resource languages with hundreds of millions of speakers who presently lack reliable MT systems, despite demonstrable need for them (Neto et al., 2020; Robinson et al., 2024). Creole language technologies have not been thoroughly researched or developed; yet several works that do exist inspired our decision to explore them (cf. our literature review in Appendix A). Although most Creoles are genealogically related to high-resource languages (*e.g.*, English, French, Spanish), previous works have had mixed success leveraging transfer learning to boost performance for Creoles (Lent et al., 2022a, 2024; Robinson et al., 2022, 2023). No prior work has conclusively determined which languages to select for successful Creole transfer, explored PEFT in this application, nor compared linguistic and non-linguistic language relations in Creole transfer language selection. These aspects combined make Creoles a viable testbed for comparing transfer criteria.

We apply a new adapter method inspired by Üstün et al.'s (2021b) approach to MT for three Creole languages. (See details in §2.) We experimented with this approach to compare a large variety of dimensions of linguistic relatedness and found that, contrary to our initial hypotheses, gains are virtually indistinguishable regardless of which transfer criterion we apply. Still more surprisingly, incorporating unrelated languages or noise to the Creole adapters yields comparable improvements. We conjecture that even untrained adaptors may assist cross-attention fine-tuning by acting as regularizers. We couple this hypothesis with analysis presented in §3.

We contribute both ① a novel regularization method combining adapter souping with cross-attention fine-tuning that surpasses baselines by 8 BLEU in some settings, and ② evidence that some of the benefit of adapters lies in regularization, independent of a wide range of linguistic information.

2 Methodology and Experiments

We follow Üstün et al. (2021a) with a two-step methodology: monolingual denoising adapter training followed by fine-tuning decoder cross-attention on small parallel bitexts. And inspired by Faisal and Anastasopoulos (2024); Wortsman et al. (2022), we induce transfer by souping Creole adapters with candidate transfer language adapters.

We focus our efforts on three Creole languages: Haitian, Papiamento, and Sango, spoken in total by roughly 18 million speakers in Haiti, the ABC Islands, and the Central African Republic, respectively.[1] This selection is primarily due to the intersection of data availability and support by NLLB-200, the MT model we apply in experiments (NLLB Team et al., 2022). Despite their unique contexts, these languages share commonalities in socio-historical development: they evolved from linguistic contact between European and African languages resulting from Western colonialism (Kouwenberg and Singler, 2009). As such, they can trace parts of their lexicon and grammar back to both European and African languages.

Selecting Languages for Transfer We examine five principled approaches to select transfer languages, which range from linguistically motivated to purely data-driven. First, we leverage language phylogeny, selecting likely Indo-European (**IE**; Rickford and McWhorter, 1997; Lefebvre, 2011) and Niger-Congo (**NC**; Hammarström et al., 2023; Mufwene, 2006; Hall, 1958; Faine, 1937) **relatives** for Haitian, Papiamento, and Sango. We also experiment with transfer between these **Creole languages**, as previous works have argued that Creole

[1] According to Ethnologue.

Dataset	Domain	Source
MADLAD	Web scrape	Kudugunta et al., 2023
NLLB-OPUS	Web scrape	NLLB Team et al., 2022
FLORES-200	Wikipedia	NLLB Team et al., 2022

Table 1: Datasets and domains used. We use MADLAD to train adapters, NLLB-OPUS to fine-tune cross-attention, and FLORES-200 to evaluate.

languages have linguistic commonalities with each other (Daval-Markussen and Bakker, 2012). We next select languages by similarity of documented typological features, calculated as **`lang2vec`** distances (Littell et al., 2017). For data-driven language selection, we determine language relatedness from similarity of NLLB-200 language embeddings (**NLLB representations**; NLLB Team et al., 2022), as well as Subword Evenness (**SuE**; Pelloni et al., 2022). For details of our language selection processes, see Appendix C.

Control experiments As an experimental control, we identify three language groups with no linguistic relation to the Creole languages in question: **Uralic** languages, **Dravidian** languages, and **CJK** (*i.e.*, Chinese, Japanese, and Korean). None of these groups share phylogeny or significant amounts of typological features with our Creole languages. The latter two also differ in terms of script. And the final group is distinct in that its languages also bear no phylogenetic relation to each other. As a final experimental control, we use a randomly initialized, untrained adapter (**`init`**). We use this untrained adapter in two ways: as a replacement for the Creole language adapter, and souped with the Creole language adapter. As Chronopoulou et al. (2023a) suggest for souping, we use the same random initialization for all untrained and trained adapters.

Experimental Details Our experiments follow the same general steps. ① We train all language adapters individually on denoising using monolingual data from MADLAD-400 (Kudugunta et al., 2023) that we preprocess and clean further.[2] ② We add the source language adapter s to the encoder and the target language adapter t to the decoder and freeze all adapter and model parameters. In adapter averaging experiments the `init` or the helper language adapters are souped with the s or t language adapters in this step as well. ③ We unfreeze the cross-attention of the decoder and use 10k parallel data from NLLB-OPUS (NLLB Team et al., 2022) to do cross-attention fine-tuning (CA-FT). ④ We evaluate MT from $s \rightarrow t$ on Flores-200 (NLLB Team et al., 2022) using mean BLEU and chrF scores. (See Table 1 for data composition.)

All experiments are conducted using a 600M-parameter distillation of the original 54B-parameter NLLB-200 model (NLLB Team et al., 2022) which has 12 encoder and 12 decoder layers, 16 attention heads, 1,024 dimensions, and a model vocabulary of over 250,000 sentence-piece tokens shared across all languages. We adopt the adapter training and hyperparameters from Üstün et al. (2021a), training bottleneck adapters on 10,000 training samples for a maximum of 100,000 training steps with early stopping, a batch size of 8, 4,000 warmup steps, and the Adam optimizer with a maximum learning rate of $2e-4$ with a gradient accumulation of 8 steps. We train CA-FT for 1 epoch with early stopping. We soup adapters with the same weight except when averaging with the `init` adapter where the weight of the Creole language to `init` adapter is $1:3$, to closer approximate the other setups where the Creole language adapter is souped with three other adapters.

3 Results and Analysis

Here we present only Creole-to-English BLEU scores. We conducted preliminary experiments using only 800 aligned fine-tuning segments on these and the inverse translation directions, and saw the same trends shown in our results here. We decided to prioritize into-English MT for our main experiments after seeing small adapter gains across the board for into-Creole. (See Appendix D for our full results suite.) Table 2 shows BLEU improvements of our approaches over both baselines: the base MT model with CA-FT, and the s and t language adapters without souping. The improvements, however, are not particularly differentiated between various transfer and control scenarios, and notably, untrained adapters match or exceed performance of trained ones.

Regularization, Not Transfer? Two facets of our results indicate that cross-lingual transfer does not explain the success of our methodology. The first is that adapter souping improves performance over the baselines, *regardless* of the criteria used to select transfer languages. The unrelated control languages (*i.e.*, Dravidian, Uralic, and CJK) achieve

[2] See Appendix B for preprocessing details.

Experiment	hat → eng	pap → eng	sag → eng
Base Model (CA-FT)	33.37	38.97	10.89
s and *t* Adapters	32.33	40.04	11.40
📻 Untrained *s* Adapter	37.07	45.01	14.91
🌳 IE Transfer	36.44	46.35	12.46
🌳 NC Transfer	36.06	46.69	12.29
❤️ Creole Transfer	35.25	46.23	12.76
📖 lang2vec	36.54	47.04	13.07
👗 NLLB Vec	35.80	46.91	12.80
⚖️ Subword Evenness	36.36	47.03	13.12
Mean Improvement	+2.70	+7.74	+1.86
📻 Untrained Souping	37.42	46.34	13.41
🎲 Uralic	37.06	47.00	13.58
🎲 CJK	36.41	47.17	13.33
🎲 Dravidian	36.55	47.27	13.27
Mean Improvement	+3.49	+7.98	+2.51

Table 2: Mean BLEU scores for Creole to English MT experiments over a diverse set of conditions for selecting transfer languages, as motivated by phylogeny (🌳), language group (❤️), typological databases (📖), model representations (👗), and tokenization (⚖️). Experimental results are also compared against control experiments with unrelated languages (🎲) and untrained adapters (📻). We highlight mean performance improvements over the base model for both transfer and controls.

FT examples:	800	10k
Base Model (CA-FT)	45.45	45.53
s and *t* Adapters	38.58	41.92
🌳 `spa + por + oci` Souping	41.87	43.74
📻 Untrained Souping	43.97	44.75

Table 3: BLEU scores for Catalan to English MT.

performance gains on par with languages selected through principled means (*e.g.*, genealogy and typology), without marked differences in BLEU or chrF scores. This holds for both the source and target language adapters in the encoder and decoder, respectively. The second is the commensurate success of the untrained adapters in averaging. Souping adapters with 'meaningful' language adapters is no better than souping with random noise, indicating that cross-lingual transfer (or any information transfer) is likely not responsible for performance gains. These findings seem more consistent with previous works on regularization, which find that adding noise can improve a model's ability to generalize and avoid overfitting (Hochreiter and Schmidhuber, 1994; Poole et al., 2014; Moradi et al., 2020) (which may be pertinent given both our limited train sets and train/test domain mismatch).

We performed multiple analyses to test our hypothesis that adapters are regularizing. First, we compared the use of untrained adapters to the use of adapters of known closely related languages. Because the close relatives of Creole languages are disputed, we turned to another relatively low resource language, Catalan, which has three close relatives also supported by NLLB-200. Spanish, Portuguese, and Occitan are closely related to Catalan by effectively every reasonable linguistic criterion. Table 3 shows different trends for Catalan (a language on which NLLB's baseline performance is notably high). In this scenario, none of the adaptor approaches improve performance over the baseline, once again indicating that the monolingual denoising adapter training on MADLAD data is not positively contributing. And once again, even with these highly attested close linguistic relationships, we see the random adapter comes nearest to recovering the baseline score in both data settings. This set of results underscores the possibility that CA-FT without randomized regularization may lead to overfitting (hence the performance drops). We also experimented souping only one or two related languages for both Catalan→English and Papiamento→English but saw no improvements.

We observed training loss, gradient norms, and dev loss for CA-FT of the experiments presented in this section. We compare untrained adapter souping (hypothesized regularization) with simple *s* and *t* adapters (unregularized), finding that the unregularized case has moderately higher gradient norms and dev loss, consistent with our hypothesis. The unregularized case also sees higher train loss, which is inconclusive regarding our hypothesis since an overfit model may be stuck in a local minimum or be overfit on a distribution varying from the CA-FT data. (See Appendix E.)

4 Conclusion

Inspired by prior success of monolingual adapters and CA-FT for low-resource MT, we corroborate prior findings that adapters can help by consistently improving BLEU scores for a set of Creole languages. However, we differ from prior research by finding no evidence of linguistic or informational transfer from adapters, even across a wide array of linguistic relationships. We hypothesize that most adapter aid in this setting is from regularization and find some analytical support for this.

Limitations

Scalability This work can only be applied to a small subset of all Creoles that are present in larger benchmarks (Lent et al., 2024; Robinson et al., 2024) due to the fact that they are low-resource languages – in certain cases, extremely so –, so there is an issue of data scarcity and a limited support by MT models. This data scarcity extends to also the African relatives of the Creoles we target in this paper. Our work is further limited by the limited coverage by typological resources. Grambank (Skirgård et al., 2023) supports only Haitian, not Papiamento and Sango, and while the lang2vec covers more languages, the coverage is limited and uneven. Despite these constraints, however, we believe our selection of Creoles (*i.e.*, Haitian, Papiamento, and Sango) are diverse enough to make high-level hypotheses about transfer learning for Creoles; these languages have different relatives (*e.g.*, inheriting large vocabulary from French, Portuguese, and Ngbandi, respectively) and are spoken across different regions (*i.e.*, the Caribbean and Africa). Moreover, our experimentation over these languages is **extensive**, with careful investigation of 6 transfer scenarios, 1 `init` baseline, and 3 principled controls. So while this work cannot scale *widely* across Creoles, we investigate transfer scenarios for each Creole *deeply*.

Domain of Data Similar to many other low-resource languages, much of the available data for Creoles comes from the religious domain (*e.g.*, Bible translations), which has a characteristic style that is typically not representative of language use in other contexts (Mielke et al., 2019). Moreover, religious domain content may not necessarily be culturally appropriate for some communities (Hershcovich et al., 2022; Mager et al., 2023). In order to be truly useful for end users, data sourced from a wide variety of domains will ultimately be a necessary component for the development of robust MT systems for Creoles. Until more such data becomes available, we believe that work at the intersection of transfer learning and domain adaptation could be greatly beneficial for Creole NLP.

Difficulty in verifying neural phenomena While we find some analytical evidence that our adapters are boosting performance by regularizing fine-tuning, it is difficult to verify this with certainty. Regularization is a mathematical phenomenon that can be demonstrated rigorously, but such involved analysis is outside the scope of this current work; hence we leave these important analytical details to future work.

Ethics Statement

As Creole languages are often marginalized, we recognize the necessity of heightened ethical scrutiny for research involving these languages. We thus engage directly with considerations from Lent et al. (2022b), Mager et al. (2023), and Bird (2020), to ensure that our work is in accordance with the needs and desires of speakers. To this end, we note that speakers of many different Creole languages actively *want* and/or *need* improved MT technology for their language. In the context of Haitian Creole, for example, MT technology is highly desirable for facilitating interactions with non-Haitian speakers (*e.g.*, the neighboring, Spanish-speaking Dominican Republic, and foreign workers at NGOs operating in Haiti). Additionally, MT has the potential to greatly improve access to information for Haitians. DeGraff et al. (2022) argues that the lack of access to education in Creole is a critical barrier to social and economic mobility in Haiti. Accordingly, reliable MT for Haitian Creole could make available high-quality educational resources from platforms such as edX, Coursera, MITx, and Udemy, for example. Thus, if MT systems could offer accurate translations into Haitian Creole, they would become transformative tools for advancing the country's development, breaking down educational barriers, and fostering greater social equity. We note similar potential for MT for speakers of Sango in Central African Republic, Chad, and Democratic Republic of the Congo and (African Development Bank Group, 2024). And while the socioeconomic context of Aruba, Bonaire, and Curacao is quite different from that of aforementioned Haiti or Central African Republic, speakers' desire for language technology for Papiamento is well documented through projects like papiamentu.ai.

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A Literature Review

NLP research for Creole languages is still in relatively early stages. Some large-scale multilingual MT systems have limited Creole language support: NLLB (NLLB Team et al., 2022) supports five Creole languages, and Google Translate[3] and Lego-MT (Yuan et al., 2023) each support eight. Other works have focused specifically on building MT for Creole languages Haitian (Robinson et al., 2022; Callison-Burch et al., 2011; Frederking et al., 1997), Singlish (Wang et al., 2017; Liu et al., 2022b), Nigerian Pidgin (Adelani et al., 2022; Ogueji and Ahia, 2019), Mauritian Creole (Dabre and Sukhoo, 2022), Sranan Tongo (Zwennicker and Stap, 2022), Guyanese Creole (Clarke et al., 2024), and Jamaican Patois (Robinson et al., 2022). Some others have focused on building Creole language-specific NLP tools outside of MT for Nigerian Pidgin (Muhammad et al., 2023, 2022; Adelani et al., 2021; Oyewusi et al., 2020; Caron et al., 2019), Jamaican Patois (Armstrong et al., 2022), and Martinican Creole (Mompelat et al., 2022).

More recently, researchers have developed large scale multi-Creole-language NLP applications. Lent et al. (2024) developed CreoleVal, a multitask benchmark for Creoles including an MT dataset and model for 28 Creole languages. Robinson et al. (2024) expanded on Lent et al.'s (2024) work, building genre-diverse MT datasets and models for 41 Creole languages of the Caribbean, Latin America, and Colonial Africa. Perhaps most pertinent to our current work are past studies into cross-lingual transfer to benefit Creole NLP. In tightly controlled experiments, Robinson et al. (2023) found indications that Haitian and Jamaican Patois' Niger-Congo relatives may be more helpful as transfer languages than their Indo-European relatives, for MT applications. Lent et al. (2022a), however, did not identify clear advantages from a language's relatedness to a Creole language in terms of its usefulness in transfer learning. Given the sparse research and even sparser positive results in the area, optimal transfer learning for Creole NLP remains an open question.

[3] https://translate.google.com/ as of 8 Feb., 2025

B Data Preprocessing

The MadLad corpus is organized in series of internet-scraped documents, which the MadLad team cross-checks for noise (Kudugunta et al., 2023). For training the monolingual adapters, we use the *MadLad-Clean* subsplit and we perform further preprocessing, since we noted after manual inspection that noise can still be present in the documents. Firstly, the MadLad documents are split into sentences using the SentenceSplitter [4] library.

For each sentence in the document, we define the following rules, if any of the rules is breached, then the given sentence is not included in the final monolingual training corpus.

1. Sentence ends with a terminal punctuation mark ('.'; '!', ' " ', '?').
2. Ratio of numerical characters with respect to all other characters is less than **0.25**.
3. Sentence starting with a bullet-point.
4. Sentence containing a *lorem ipsum* substring.
5. Ratio of stop-words with respect to all other tokens is less than **0.6** using whitespace tokenization.
6. Ratio of token count that does not contain alpha characters with respect to total token count is less than **0.25**, using whitespace tokenization.
7. Sentence length is less than **14** and more than **300** characters.
8. Ratio of capitalized characters is less than **0.2**.
9. Ratio of alpha characters with respect to all other characters is more than **0.6**.
10. Language identification does not match the expected language. For this we employ FastText model [5].

This preprocessing removed approximately 20-23% of sentences, depending on the language. The same preprocessing steps were also applied to NLLB parallel data used for cross-attention fine-tuning, where preprocessing removed approximately 30-50% of the sentences.

C Selecting Languages for Transfer

Table 4 displays the different transfer languages we selected in our experiments. Below we detail our processes of selecting them.

[4] https://github.com/mediacloud/sentence-splitter

[5] https://huggingface.co/facebook/fasttext-language-identification

Condition	Language
Haitian	
Creole Language	hat
IE Relative	fr, es, en
NC Relative	ig, ee, fon
`lang2vec`	grc, oc, fo
NLLB	pap, epo, ltz
Papiamento	
Creole Language	pap
IE Relative	nl, pt, es
NC Relative	kg, ak, fon
`lang2vec`	suz, sr, ug
NLLB	hat, war, tgl
Sango	
Creole Language	sag
IE Relative	fr
NC Relative	kbp
`lang2vec`	ig, yo, ca
NLLB	kg, tsn, lin
Subword Evenness	kat, rus, zsm
Controls	
Uralic	fin, est, hun
CJK	zho, jap, kor
Dravidian	te, ta, ka
Other Settings	
Hungarian	
`lang2vec`	fin, tur, als
NLLB	fin, bul, hrv
Yoruba	
`lang2vec`	tha, ibo, fon
NLLB	ibo, hau, xho

Table 4: All languages used for the various experimental settings.

C.1 Phylogeny-based Transfer

Faisal and Anastasopoulos (2022) followed principles of language phylogenetics to inform their transfer learning experiments, training adapters on combined Indo-European data, and for subfamilies of Indo-European such as Germanic, Romance, Indo-Aryan, etc. To translate for French, for example, they stacked a French adapter on top of a Romance and then an Indo-European adapter at inference time. Creole phylogeny is heavily disputed, however (Kouwenberg and Singler, 2009). Linguists have debated whether Atlantic Creole languages such as Haitian and Papiamento are best understood as Niger-Congo (NC) languages with extensive Indo-European (IE) lexical borrowings (Lefebvre, 2011; Rickford and McWhorter, 1997), or as IE languages that developed with strong NC influence (Faine, 1937; Hall, 1958; Hammarström et al., 2023; Mufwene, 2006). Considering this lack of consensus, we select two groups of related languages for each Creole language: IE relatives and NC relatives. In this work, we made a selection based on APICS (Michaelis et al., 2013) and linguistic references.

C.2 Typological Feature-based Transfer

Information about the typological features of languages is typically documented in databases such as APICS (Michaelis et al., 2013), WALS (Dryer and Haspelmath, 2013), and Grambank (Skirgård et al., 2023). The `lang2vec` library (Littell et al., 2017) assembles typological feature vectors from different sources, enabling measurement of holistic typological distance between languages. The drawbacks of `lang2vec` include the lack of equal coverage of languages in terms of the number of features, and the coarseness of the linguistic properties (Ponti et al., 2019). To supplant the lack of coverage, we use `data2lang2vec` which uses textual data to inform feature-prediction of the missing features (Amirzadeh et al., 2025).

C.3 Data-driven Language Selection Methods

NLLB Language Embeddings Typological similarity between languages can also be derived in a data-driven manner, *e.g.*, from language embeddings. To derive the similarity between languages this way, we utilize the language embeddings of the distilled NLLB-200 model which implicitly contain much information with regards to the various languages especially with regards to morphosyntax and vocabulary.

Subword Evenness Introduced by Pelloni et al. (2022), Subword Evenness (SuE) is a metric quantifying the differences in the length of subword units for each individual language. It presents another data-driven approach to transfer language selection. In this metric, lower SuE scores have been demonstrated as a good predictor of transfer success to various low-resource languages. Thus, we select the 5 languages in the MADLAD dataset with the lowest SuE angles as calculated when using a BPE tokenizer, see Table 5.

Language Code	Language Code	Script	SuE Angle
≺azb	South Azerbaijani	Arabic	66.72
kat	Georgian	Georgian	72.57
zsm	Malay	Latin	79.80
≺kas	Kashmiri	Devanagari	80.79
rus	Russian	Cyrillic	81.21

Table 5: MADLAD Languages with the lowest SuE angles calculated using a BPE tokenizer. We note that MADLAD only has Norther Azerbaijani (Latin script), and not South Azerbaijani (Arabic script). Similarly, Kashmiri is only in MADLAD with the Arabic script, not with Devanagari.

C.4 Control Experiments with Unrelated Languages

To determine whether improvements over Creoles are truly the result of successful cross-lingual transfer, we compare the previously described methods against an equivalent adapter approach using entirely unrelated languages. We use three different sets of such unrelated languages. The first set is made up of the **Uralic languages** Finnish, Estonian, and Hungarian, which are written with the Latin script but have no phylogenetic relation to Haitian, Papiamento, and Sango. The second set consists of Chinese, Japanese, and Korean – so-called **CJK languages** –, which belong to three distinct language families and are written with three distinct scripts having little to no overlap with Atlantic and African Creole languages. The third group consists of the **Dravidian languages** Tamil, Telugu, and Kannada, which are closely related to each other but also written with three distinct scripts, and again are unrelated to the Creole languages in terms of phylogeny.

D Full Results

See Table 6 for our results translating between English and Creole languages using only 800 fine-tuning parallel segments.

E Training Loss

Table 7 compares loss when evaluating on a small subset of the NLLB CA-FT training data. Figure 2 indicates that the adapters we train learn successfully from the MADLAD-400 data on a selection from all language adapters.

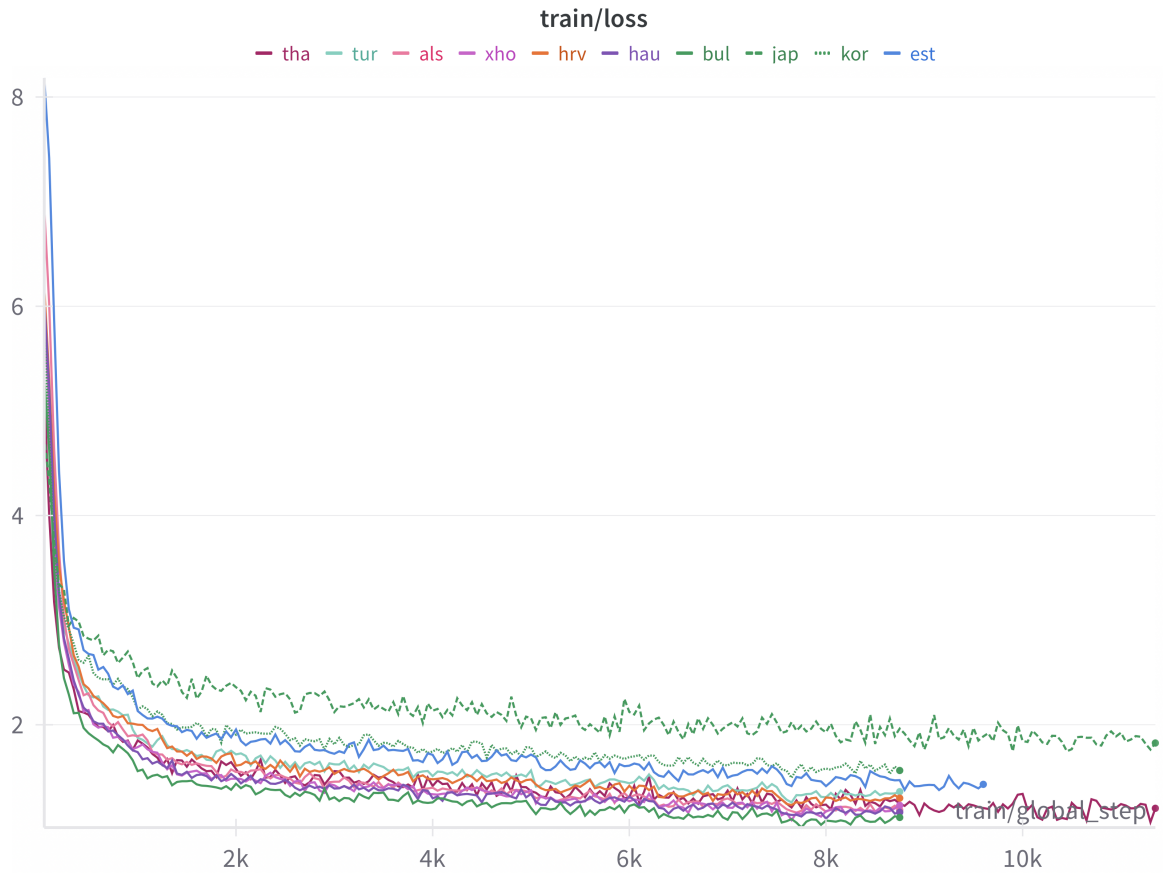


Figure 2: Training loss curves for a subset of adapter training experiments.

Experiments	en → hat	en → pap	en → sag	hat → en	pap → en	sag → en
Baselines						
dNLLB-200 w/ CA-FT	23.59	23.56	11.16	33.36	39.05	10.62
Language Adapters						
Source + Target LA	23.55	26.82	11.90	31.18	38.23	12.19
Phylogeny-Inspired Adaptation						
Target + IE Relatives	24.01	29.35	11.94	35.24	45.48	13.51
Target + NC Relatives	23.98	29.71	11.93	34.96	46.13	13.03
Typological Adaptation						
`lang2vec`	**24.28**	**30.21**	11.71	**35.63**	**46.95**	13.79
Target + Data-Driven (NLLB)	24.09	30.17	**12.01**	34.75	46.09	13.48
Miscellaneous						
Creole Transfer	23.99	29.87	11.71	34.24	44.75	13.48
Subword Evenness	24.23	29.73	11.80	35.20	46.20	**14.11**
Largest Improvement from dNLLB	+0.69	+6.65	+0.85	+2.27	+7.90	+3.49
Largest Improvement from Üstün	+0.73	+3.39	+0.11	+4.45	+8.63	+1.92
Control						
Uralic	24.31	30.08	11.44	**36.01**	46.36	**15.03**
CJK	23.95	29.18	11.60	35.64	46.91	14.71
Dravidian	24.21	29.17	11.69	35.51	46.66	14.45
Random (same init) (1 : 1)	24.21	28.89	11.86	36.07	46.03	14.79
Random (same init) (1 : 3)	24.10	29.65	11.55	37.10	46.66	15.39
Only random (same init)	-	-	-	36.84	45.01	14.90

Table 6: Mean BLEU scores on Flores-200 (after 1 run). Results follow adapter training defaults used in the Üstün et al. (2021a) paper. Encoder and decoder adapters are trained together and trained further using cross-attention fine-tuning (CA-FT), and helper and target language adapters are averaged (souped) together. **These scores represent preliminary results, where significantly less data (800 segments) were used to do CA-FT, and thus these results do not correspond to Table 2.**

Experiment	hat → eng	pap → eng	sag → eng
s and t Adapters	0.594	1.106	1.908
Untrained s Adapter	**0.476**	**0.917**	**1.796**

Table 7: Validation loss when evaluating on the CA-FT training data. Averaging with the weights with the same initialization results in lower loss values.

Experiments	en → hu	en → yo	hu → en	yo → en
dNLLB-200	22.59	8.85	29.51	18.82
`lang2vec`	21.85	**8.51**	33.38	19.27
NLLB	21.75	8.45	33.31	19.14
SuE	**21.90**	8.37	33.47	19.56
Random (init) (1 : 3)	21.88	8.47	**33.97**	**20.70**
Difference	−0.69	−0.34	+4.46	+1.88

Table 8: Mean BLEU scores with our methodology, as applied to Yoruba and Hungarian. **These scores represent preliminary results, where significantly less data (800 segments) were used to do CA-FT, and thus not directly comparable to Table 2.**